ASD20
Professional conduct & Ethics

Johan Bontes

Other professions

E.g.: charted accountants

Accreditation differs per country, but me:

- 4 Years undergrad
- 3 Years postgrad (comparable to Master's)
- 3 Years apprenticeship under charted accountant (concurrent with postgrad)

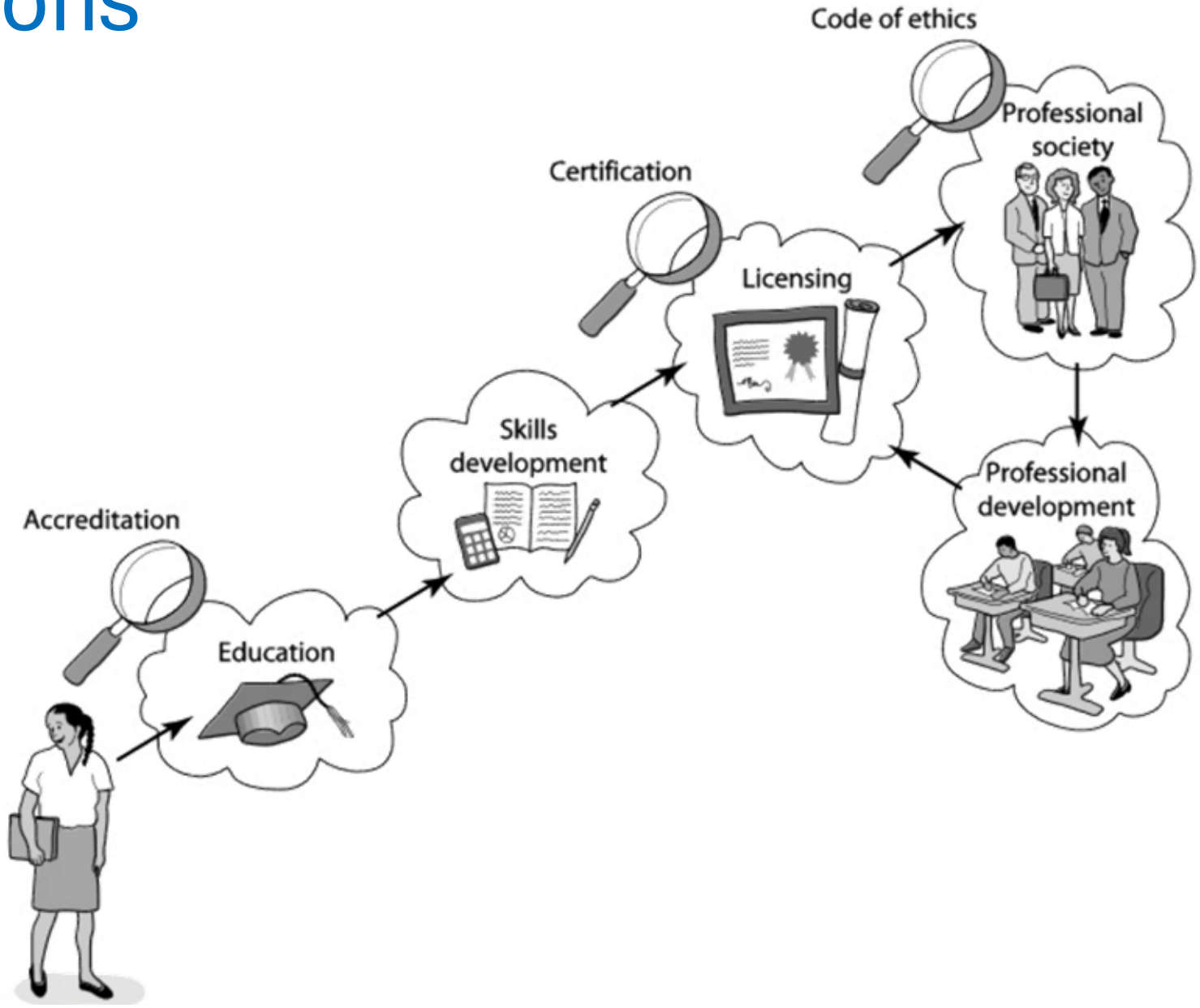
Total: 7 years university level

- Compulsory membership of professional oversight body
- Permanent education requirements
- Disciplinary hearings on violation of code of ethics
- Subject to legal requirements codified in law

Other professions

Similar for

- Medical doctors
- Architects
- Lawyers



Software developers

-- nothing --

nullptr exception

- No certification or licensing
- No formal education requirements
- No apprenticeship
- No compulsory membership of professional body
- No disciplinary mechanisms
- Voluntary code of ethics

ACM Code of ethics (2018)

1. GENERAL ETHICAL PRINCIPLES.

1.1 Contribute to society and to human well-being, acknowledging that all people are stakeholders in computing.

1.2 Avoid harm.

1.3 Be honest and trustworthy.

1.4 Be fair and take action not to discriminate.

1.5 Respect the work required to produce new ideas, inventions, creative works, and computing artifacts.

1.6 Respect privacy.

1.7 Honor confidentiality.

ACM Code of ethics

2. PROFESSIONAL RESPONSIBILITIES.

2.1 Strive to achieve high quality in both the processes and products of professional work.

2.2 Maintain high standards of professional competence, conduct, and ethical practice.

2.3 Know and respect existing rules pertaining to professional work.

2.4 Accept and provide appropriate professional review.

2.5 Give comprehensive and thorough evaluations of computer systems and their impacts, including analysis of possible risks.

2.6 Perform work only in areas of competence.

2.7 Foster public awareness and understanding of computing, related technologies, and their consequences.

2.8 Access computing and communication resources only when authorized or when compelled by the public good.

2.9 Design and implement systems that are robustly and usably secure.

ACM Code of ethics

3. PROFESSIONAL LEADERSHIP PRINCIPLES.

3.1 Ensure that the public good is the central concern during all professional computing work.

3.2 Articulate, encourage acceptance of, and evaluate fulfilment of social responsibilities by members of the organization or group.

3.3 Manage personnel and resources to enhance the quality of working life.

3.4 Articulate, apply, and support policies and processes that reflect the principles of the Code.

3.5 Create opportunities for members of the organization or group to grow as professionals.

3.6 Use care when modifying or retiring systems.

3.7 Recognize and take special care of systems that become integrated into the infrastructure of society.

ACM Code of ethics

4. COMPLIANCE WITH THE CODE.

4.1 Uphold, promote, and respect the principles of the Code.

4.2 Treat violations of the Code as inconsistent with membership in the ACM.

IEEE Code of ethics (2020)

I. To uphold the highest standards of integrity, responsible behavior, and ethical conduct in professional activities.

1. to hold paramount the safety, health, and welfare of the public, to strive to comply with ethical design and sustainable development practices, to protect the privacy of others, and to disclose promptly factors that might endanger the public or the environment;

2. to improve the understanding by individuals and society of the capabilities and societal implications of conventional and emerging technologies, including intelligent systems;

3. to avoid real or perceived conflicts of interest whenever possible, and to disclose them to affected parties when they do exist;

IEEE Code of ethics

4. to avoid unlawful conduct in professional activities, and to reject bribery in all its forms;

5. to seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, to be honest and realistic in stating claims or estimates based on available data, and to credit properly the contributions of others;

6. to maintain and improve our technical competence and to undertake technological tasks for others only if qualified by training or experience, or after full disclosure of pertinent limitations;

IEEE Code of conduct

7. to treat all persons fairly and with respect, and to not engage in discrimination based on characteristics such as race, religion, gender, disability, age, national origin, sexual orientation, gender identity, or gender expression;

8. to not engage in harassment of any kind, including sexual harassment or bullying behavior;

9. to avoid injuring others, their property, reputation, or employment by false or malicious actions, rumors or any other verbal or physical abuses;

10. to support colleagues and co-workers in following this code of ethics, to strive to ensure the code is upheld, and to not retaliate against individuals reporting a violation.

Do not lawyerize code of ethics

American legal system:

- Exceptionally detailed
- Every possible eventuality captured in legal clauses
  - Feels secure, but encourages ‘gaming the system’
    - If it’s not explicitly forbidden, then it must be allowed
  - Forces ever increasing level of detail

European legal system:

- Mostly derived from Roman Law (449 BC – 526 AD)
  - Principle based
    - fraus legis: fraudulent application of the letter of the law
    - acts are legal only if compliant with the intent of the law

Licensing of qualification?

ACM opposes it

- Premature
  - *seems like a weak argument*
  - Computing pervades all aspects of modern life

- Would not be effective in improving quality and reliability of software
  - *debatable, because Centers of excellence aside*
  - Certified minimal level of training and experience would improve Q & R
  - Enforceable codes of ethics and professionalism
    - Would give public means to hold professionals to account
    - Will obviously not stop violations from happening

Boeing MCAS

Was the behavior by the programmers ethical?

Boeing MCAS

Was the behavior by the programmers ethical?

- Harm?
- Competence?
- High quality?
- Honesty?
- Lawful conduct (FAA compliance)?
- Care when modifying system?

Generative AI

Example: Meta LLM

- Downloaded every book on pirate site libgen.is to train LLM model
- Using bittorrent
  - claims this is ‘legal’ because
    they did not enable sharing of torrented data (leeching only)
  - clear violation of ACM/IEEE code of ethics
  - software developers have no legally enforceable code of ethics
    - unlike accountants/lawyers/architects

- Generative AI threatens to put content creators out of work
  - artists/writers/film makers
  - and yes, programmers too

Intellectual property

Copyright

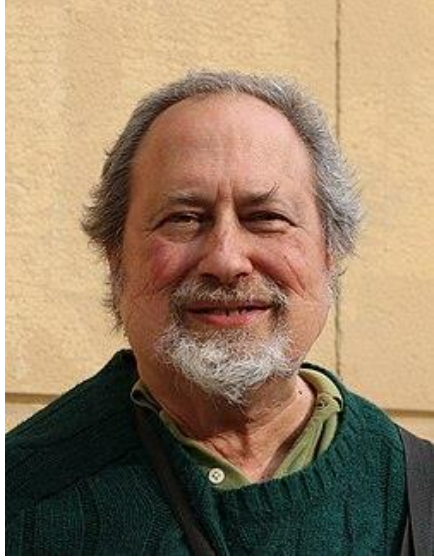
Patents

Licensing terms

International framework

- Not all countries respect IP
  - e.g.: Russia disregards copyright entirely

Richard Stallmann



Copyleft (aka GPL license)

uses copyright law to preserve the right to use, modify, and distribute free software

Member of FSF (Free Software Foundation)

copyright/patents/
digital restrictions management (DRM)
Are misused to lock information away from users/programmers indefinitely



Defective by design

Advocates for boycott, not piracy

Pirate bay/libgen/sci-hub



Sci-hub: Alexandra Elbakyan

- Paused since 2020 due to Indian lawsuit.

"The goal of Sci-Hub is to provide free and unrestricted access to all scientific knowledge"

Elbakyan: "legal" under the right to science and culture under Article 27 of the Universal Declaration of Human Rights

Others: Sci-Hub = moral imperative (civil disobedience)

IP Law should be changed, not Sci-Hub banned

Legal vs philosophical ethics

Law is a balance of interests

- does not mean balance is just or fair
- 'Legal' does not mean good
  - Laws can be 'captured' by special interests
    - Apartheid laws were legal
    - NSDAP (Nazi party) complied fully with German law

My take:

- If you join an organisation
  - either subscribe to its core tenets
  - or resign
- power to change is very limited

Common software licenses

Open source

- GPL
- LGPL
- Apache
- MIT

Proprietary

- Garubi license

Plagiarism & authorship

Unethical for scholars

- Reputation build on original research
- Undermines meritocracy
- Dishonest

Unethical for developers

- Violates intellectual property
- Makes it impossible to track source of code
  - Cannot research its limitations/issues
- Makes you look like a fraud
- Also: risk: p:H c:H => avoid